Math 162
Bonus problem solutions

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1. (a) Assuming that $x \neq 0$, we can integrate by parts to obtain

$$\int_0^{\frac{\pi}{2}} \cos^n t \cos 2xt \, \mathrm{d}t = \cos^n t \frac{\sin 2xt}{2x}\bigg|_{t=0}^{t=\frac{\pi}{2}} + \frac{n}{2x} \int_0^{\frac{\pi}{2}} \cos^{n-1} t \sin t \sin 2xt \, \mathrm{d}t$$
$$= \frac{n}{2x} \int_0^{\frac{\pi}{2}} \cos^{n-1} t \sin t \sin 2xt \, \mathrm{d}t \, .$$

Integrate by parts once more:

$$\int_0^{\frac{\pi}{2}} \cos^{n-1} t \sin t \sin 2xt \, \mathrm{d}t = -\cos^{n-1} t \sin t \frac{\cos 2xt}{2x}\bigg|_{t=0}^{t=\frac{\pi}{2}}$$
$$+ \frac{1}{2x} \int_0^{\frac{\pi}{2}} (-(n-1)\cos^{n-2} t \sin^2 t + \cos^n t) \cos 2xt \, \mathrm{d}t$$
$$= -\frac{n-1}{2x} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \cos 2xt \, \mathrm{d}t + \frac{n}{2x} \int_0^{\frac{\pi}{2}} \cos^n t \cos 2xt \, \mathrm{d}t \, .$$

Combining these yields

$$\int_0^{\frac{\pi}{2}} \cos^n t \cos 2xt \, \mathrm{d}t = -\frac{n(n-1)}{4x^2} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \cos 2xt \, \mathrm{d}t + \frac{n^2}{4x^2} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \cos 2xt \, \mathrm{d}t,$$

and if we assume that $x \neq \pm\frac{n}{2}$ this can be rearranged as

$$\int_0^{\frac{\pi}{2}} \cos^n t \cos 2xt \, \mathrm{d}t = -\frac{n(n-1)}{4x^2} \left(1 - \frac{n^2}{4x^2}\right)^{-1} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \cos 2xt \, \mathrm{d}t \, .$$

Straightforward algebra confirms that

$$-\frac{n(n-1)}{4x^2} \left(1 - \frac{n^2}{4x^2}\right)^{-1} = \frac{n-1}{n} \left(1 - \frac{4x^2}{n^2}\right)^{-1},$$

from which we conclude that

$$\int_0^{\frac{\pi}{2}} \cos^n t \cos 2xt \, \mathrm{d}t = \frac{n-1}{n} \left(1 - \frac{4x^2}{n^2}\right)^{-1} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \cos 2xt \, \mathrm{d}t$$

for $x \neq 0, \pm\frac{n}{2}$. Finally, notice that when $x = 0$ this identity reduces to

$$\int_0^{\frac{\pi}{2}} \cos^n t \, \mathrm{d}t = \frac{n-1}{n} \int_0^{\frac{\pi}{2}} \cos^{n-2} t \, \mathrm{d}t,$$

which we proved in #5(a) on Problem Set 9.

(b) Performing a simple induction using the identity from (a) yields

$$\int_0^{\frac{\pi}{2}} \cos^{2n} t \cos 2xt \,\mathrm{d}t = \left(\prod_{k=1}^{n} \frac{2k-1}{2k}\left(1-\frac{x^2}{k^2}\right)^{-1}\right)\int_0^{\frac{\pi}{2}} \cos 2xt \,\mathrm{d}t$$
$$= \left(\prod_{k=1}^{n} \frac{2k-1}{2k}\left(1-\frac{x^2}{k^2}\right)^{-1}\right)\frac{\sin \pi x}{x}$$

for $n \in \mathbb{N}$ and $x \neq 0, \pm 1, \cdots, \pm n$. Combine this with the identity

$$\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t = \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}$$

from #5(a) on Problem Set 9 and rearrange to obtain

$$\sin \pi x = \pi x \prod_{k=1}^{n}\left(1-\frac{x^2}{k^2}\right)\frac{\int_0^{\frac{\pi}{2}} \cos^{2n} t \cos 2xt \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t},$$

again for $x \neq 0, \pm 1, \cdots, \pm n$. Now observe that this equation also holds for the excluded values of x: the left side vanishes for all $x \in \mathbb{Z}$, and on the right side the product

$$x \prod_{k=1}^{n}\left(1-\frac{x^2}{k^2}\right)$$

vanishes for $x = 0, \pm 1, \cdots, \pm n$.

(c) Similarly to #5(b) on Problem Set 9, we claim that

$$\frac{2n}{2n+1} \leq \frac{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n-1} t \,\mathrm{d}t} \leq 1$$

for any $n \in \mathbb{N}$. Namely, this follows from the inequalities

$$\frac{2n}{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n-1} t \,\mathrm{d}t = \int_0^{\frac{\pi}{2}} \cos^{2n+1} t \,\mathrm{d}t \leq \int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t \leq \int_0^{\frac{\pi}{2}} \cos^{2n-1} t \,\mathrm{d}t\,.$$

Here the equation was proved in #5(a) on Problem Set 9, and the inequalities result from the observation that $0 \leq \cos t \leq 1$ for $0 \leq t \leq \frac{\pi}{2}$.

Next, we use the product formulas from #5(a) on Problem Set 9 to see that

$$\frac{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n-1} t \,\mathrm{d}t} = \frac{\pi}{2}\prod_{k=1}^{n}\frac{2k-1}{2k}\left(\prod_{k=1}^{n-1}\frac{2k}{2k-1}\right) = \frac{2}{\pi}\left(\prod_{k=1}^{n}\frac{2k-1}{2k}\right)^2 2n.$$

Combining this with the above inequalities, we obtain

$$\frac{2n}{2n+1} \leq \frac{2}{\pi}\left(\prod_{k=1}^{n}\frac{2k-1}{2k}\right)^2 2n \leq 1,$$

or after rearrangement

$$\frac{\pi}{4}\frac{1}{n+\frac{1}{2}} \leq \left(\prod_{k=1}^{n}\frac{2k-1}{2k}\right)^2 \leq \frac{\pi}{4}\frac{1}{n}.$$

Take the square root to complete the proof.

(d) Following the hint, we claim that $\cos t \leq \frac{1}{\sqrt{1+t^2}}$ for $0 \leq t \leq \frac{\pi}{2}$. One way to do this is to first observe that $t \leq \tan t$ for $0 \leq t < \frac{\pi}{2}$. Namely, for such t we have $1 \leq \sec^2 t$ and hence

$$t = \int_0^t \mathrm{d}s \leq \int_0^t \sec^2 s \,\mathrm{d}s = \tan t.$$

Then we deduce that

$$t^2 \leq \tan^2 x = \sec^2 t - 1,$$

which can then be rearranged into the inequality of the claim (the latter is obvious when $x = \frac{\pi}{2}$). Next, observe that

$$\begin{aligned}
\left| \int_0^{\frac{\pi}{2}} f(t) \cos^{2n} t \,\mathrm{d}t - f(0) \int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t \right| &= \left| \int_0^{\frac{\pi}{2}} (f(t) - f(0)) \cos^{2n} t \,\mathrm{d}t \right| \\
&\leq \int_0^{\frac{\pi}{2}} |f(t) - f(0)| \cos^{2n} t \,\mathrm{d}t \\
&\leq M \int_0^{\frac{\pi}{2}} t \cos^{2n} t \,\mathrm{d}t \\
&\leq M \int_0^{\frac{\pi}{2}} \frac{t}{(1+t^2)^n} \,\mathrm{d}t,
\end{aligned}$$

where in the last step we used the inequality from the claim. We have

$$\int_0^{\frac{\pi}{2}} \frac{t}{(1+t^2)^n} \,\mathrm{d}t = -\frac{1}{2(n-1)} \frac{1}{(1+t^2)^{n-1}} \bigg|_0^{\frac{\pi}{2}} < \frac{1}{2(n-1)}$$

for $n > 1$, from which we obtain

$$\left| \int_0^{\frac{\pi}{2}} f(t) \cos^{2n} t \,\mathrm{d}t - f(0) \int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t \right| < \frac{M}{2(n-1)}.$$

Using the lower bound from part (c), we deduce that

$$\left| \frac{\int_0^{\frac{\pi}{2}} f(t) \cos^{2n} t \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t} - f(0) \right| = \frac{\left| \int_0^{\frac{\pi}{2}} f(t) \cos^{2n} t \,\mathrm{d}t - f(0) \int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t \right|}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t} < \frac{2}{\sqrt{\pi}} \sqrt{n + \frac{1}{2}} \cdot \frac{M}{2(n-1)}.$$

The right side tends to zero as $n \to \infty$, for instance because

$$\lim_{n \to \infty} \frac{\sqrt{n + \frac{1}{2}}}{n-1} = \lim_{n \to \infty} \frac{\sqrt{1 + \frac{1}{2n}}}{\sqrt{n} - \frac{1}{\sqrt{n}}} = 0.$$

So we are done by the squeeze theorem.

(e) Let us first show that for any $x \in \mathbb{R}$, we have

$$\lim_{n \to \infty} \frac{\int_0^{\frac{\pi}{2}} \cos^{2n} t \cos 2xt \,\mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \,\mathrm{d}t} = 1.$$

Let $f : [0, \frac{\pi}{2}] \to \mathbb{R}$ be defined by $f(t) = \cos 2xt$. Since $f(0) = 1$, it suffices to prove that f satisfies the hypotheses of part (d). Certainly f is integrable, being continuous. For any $t > 0$, by the mean value theorem implies that there exists $t_0 \in [0, t]$ such that $f'(t_0) = \frac{f(t) - f(0)}{t}$. Thus we have

$$|f(t) - f(0)| = |f'(t_0)| t = |-2x \sin 2xt_0| t \leq 2|x| t$$

as needed.

It suffices to prove the first product formula, which implies the second after substituting $\frac{x}{\pi}$ for x. Let us first consider the case where $x \notin \mathbb{Z}$. Then all of the terms appearing in the product formula are nonzero, and $\sin \pi x \neq 0$, so it suffices to observe that

$$\lim_{n\to\infty} \pi x \prod_{k=1}^{n} \left(1 - \frac{x^2}{k^2}\right) = \lim_{n\to\infty} \left(\frac{\int_0^{\frac{\pi}{2}} \cos^{2n} t \cos 2xt \, \mathrm{d}t}{\int_0^{\frac{\pi}{2}} \cos^{2n} t \, \mathrm{d}t} \right)^{-1} \sin \pi x = \sin \pi x$$

by (b) and the limit we just computed.

If $x = N \in \mathbb{Z}$, then we have $1 - \frac{x^2}{k^2} \neq 0$ for all $k \in \mathbb{N}$ such that $k \neq |N|$, so to prove that the infinite product converges to zero in this case, we must show that

$$\prod_{k=N+1}^{\infty} \left(1 - \frac{x^2}{k^2}\right)$$

converges to a nonzero value. Since the series

$$\sum_{k=1}^{\infty} \frac{x^2}{k^2} = x^2 \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges, it suffices to prove the following claim: if $0 \leq b_n < 1$ for all $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} b_n$ converges, then the product

$$\prod_{n=1}^{\infty} (1 - b_n)$$

converges to a nonzero value (note the similarity with #4(c) on Problem Set 8). By #4(b) on the same problem set, it is enough to show that $\sum_{n=1}^{\infty} \log(1 - b_n)$ converges. For this, observe that

$$\lim_{x\to 0^+} \frac{-\log(1-x)}{x} = \lim_{x\to 0^+} \frac{1}{1-x} = 1$$

by L'Hôpital's rule, so there exists $\epsilon > 0$ such that $0 \leq x < \epsilon$ implies that

$$|\log(1-x)| = -\log(1-x) \leq 2x.$$

Since $b_n < \epsilon$ for sufficiently large n, the direct comparison test implies that $\sum_{n=1}^{\infty} \log(1-b_n)$ converges absolutely.

2. (a) Begin by observing that

$$\lim_{n\to\infty} \frac{n^2}{n^2 - r^2} = 1,$$

so there exists $N \in \mathbb{N}$ such that $n > N$ implies that $\frac{1}{n^2 - r^2} \leq \frac{2}{n^2}$. Making N larger if necessary, we can assume that $N > r$, whence

$$\left| \frac{1}{x^2 - n^2} \right| = \frac{1}{n^2 - x^2} \leq \frac{1}{n^2 - r^2} \leq \frac{2}{n^2}$$

for all $n > N$ and all $x \in [-r, r]$. Since the series

$$\sum_{n=1}^{\infty} \frac{2}{n^2} = 2 \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, the Weierstrass M-test therefore implies that the series of functions in question converges absolutely uniformly on $[-r, r] \cap A$ (the set $\mathbb{Z} \setminus \{0\}$ is excluded simply to avoid dividing by zero).

(b) Fix $x \in A$ and choose $N \in \mathbb{N}$ such that $N > |x|$. Then we have $f(x) = g(x)h(x)$ where

$$g(x) = \prod_{n=1}^{N} \left(1 - \frac{x^2}{n^2}\right)$$

and

$$h(x) = \prod_{n=N+1}^{\infty} \left(1 - \frac{x^2}{n^2}\right).$$

Applying the product rule, we have

$$g'(x) = \sum_{n=1}^{N} \frac{-2x}{n^2} \frac{g(x)}{1 - x^2/n^2} = 2xg(x) \sum_{n=1}^{N} \frac{1}{x^2 - n^2}.$$

Since $n > N > |x|$ implies that $1 - \frac{x^2}{n^2} > 0$, we can consider

$$\log h(x) = \sum_{n=N+1}^{\infty} \log \left(1 - \frac{x^2}{n^2}\right).$$

Since the functions

$$\frac{\mathrm{d}}{\mathrm{d}x} \left(1 - \frac{x^2}{n^2}\right) = \frac{2x}{x^2 - n^2}$$

are continuous, part (a) and the theorem on differentiation and uniform convergence imply that

$$\frac{h'(x)}{h(x)} = \frac{\mathrm{d}}{\mathrm{d}x} \log h(x) = \frac{\mathrm{d}}{\mathrm{d}x} \sum_{n=N+1}^{\infty} \log \left(1 - \frac{x^2}{n^2}\right) = 2x \sum_{n=N+1}^{\infty} \frac{1}{x^2 - n^2}.$$

Combining the above calculations, we obtain

$$\begin{aligned} f'(x) &= g'(x)h(x) + g(x)h'(x) \\ &= 2xg(x) \left(\sum_{n=1}^{N} \frac{1}{x^2 - n^2}\right) h(x) + g(x) \cdot 2xh(x) \sum_{n=N+1}^{\infty} \frac{1}{x^2 - n^2} \\ &= 2xf(x) \sum_{n=1}^{\infty} \frac{1}{x^2 - n^2} \end{aligned}$$

as desired.

(c) Similarly to part (a), one shows that for any $r > 0$, there exists $N \in \mathbb{N}$ such that $n > N$ implies that

$$\frac{1}{(x^2 - n^2)^2} \leq \frac{2}{n^4}$$

for all $x \in [-r, r]$. Then the Weierstrass M-test implies that the series

$$\sum_{n=1}^{\infty} \frac{1}{(x^2 - n^2)^2}$$

converges uniformly on $[-r, r] \cap A$. Since the functions $\frac{1}{(x^2-n^2)^2}$ are continuous, it follows that

$$\frac{\mathrm{d}}{\mathrm{d}x} \sum_{n=1}^{\infty} \frac{1}{x^2 - n^2} = -2x \sum_{n=1}^{\infty} \frac{1}{(x^2 - n^2)^2}$$

for any $x \in \mathbb{A}$. The claim now follows immediately by differentiating the identity in part (b).

(d) Clearly $f(0) = 1$. Thus when we set $x = 0$ in the identity from part (c), we obtain

$$f''(0) = -2 \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

On the other hand, for all $x \in \mathbb{R}$ we have

$$\begin{aligned} f(x) &= \begin{cases} \frac{\sin \pi x}{\pi x} & \text{for } x \neq 0 \\ 1 & \text{for } x = 0 \end{cases} \\ &= 1 - \frac{\pi^2 x^2}{3!} + \frac{\pi^4 x^4}{5!} - \cdots . \end{aligned}$$

Here the first equation follows from #1(f) as noted in part (b), and the second equation follows from the Taylor series expansion of $\sin x$ at $x = 0$. We conclude that

$$f''(0) = -\frac{\pi^2}{3}$$

and hence

$$-2 \sum_{n=1}^{\infty} \frac{1}{n^2} = -\frac{\pi^2}{3},$$

which is equivalent to the desired identity.